UNIVERSITI TUNKU ABDUL RAHMAN

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UCCD 1004 PROGRAMMING CONCEPTS AND PRACTICES

ASSIGNMENT 2

Group 52			
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ii. Task Division

	Name	Modules	Description	*A2 Contribution (Overall, %)	
1.	WONG PEI KEI	Add/Delete	void add(); / void mbSignup(); / bool testNum(char[]. int); void dlt(); / void mbcancelacc(int, bool&);	1	100
		View	void read(); / void mbbuyticket(int, int, double&); void mbwallet(double &);	2	100
		Check in/ Check out	void mbLogin(int&); / void mbMenu();	3	100
2.	BERNARD GOH PEK HONG	Time/Date	void mbbuyticket(int, int, double&);	1	100

		Seat	void mbbuyticket(int, int. double&);
			void print_seating_chart();
			bool reserve_seat(int, int);
				2	100

		Payment	void mbbuyticket(int, int, double&); double calculate_total_cost(int, int);	3	100
3.	ENG ZI YEE	Check in	void mbLogin(int&); / void mbMenu();	1	100
		Check out	void mbLogin(int&); / void mbMenu();	2	100
		-	-	3	0

* Depends on the evaluation of the markers as well;

iii. Objective

Add / Delete module

- Admin mode (add/delete movie using txt file to save record)

- Member mode (register new user/delete current user account using txt file to save record)

View module

- Admin mode (to view movie list whether any movie add or delete from user input)

- Member mode (to view movie list whether any movie have chosen by member to buy their ticket)

- Wallet section (to view member balance whether any top up service upload or member buy ticket deducted from payment and updated)

Check in / Check out module

- Member mode (user login check whether exist or not, if exist, proceed to member menu)

- Member mode (user proceed to logout, auto logout)

Date / Time module

- Member buy ticket can choose a specific date and time

Seat module

- Member can choose seat by rows to columns. If seat reserved, it wouldn't allow to choose

Payment module

- Payment should deduct from member wallet, if wallet not enough money, auto cancel transaction. Else, proceed.

- After member made their payment, ticket will be print out, member should have their QR code to scan.

iv. Pseudocode

```
INCLUDE iostream library
INCLUDE string library
INCLUDE fstream library
INCLUDE iomanip library
INCLUDE cctype library
INCLUDE cstring library
DEFINE the CRT_SECURE_NO_WARNINGS macro
USING namespace std

DECLARE Constant Integer arraySize = 1000
DECLARE Constant Integer pwSize = 9
DECLARE Constant Integer rows = 10
DECLARE Constant Integer columns = 20
DECLARE Integer seats[rows][columns] = { 0 }

DEFINE Struct movieType
        DECLARE String name
        DECLARE Double price

DEFINE Struct memberSign
        DECLARE String name
        DECLARE Character password[pwSize]

FUNCTION Void admb()
FUNCTION Void admMenu(int&)
FUNCTION Void read()
FUNCTION Void add()
FUNCTION Void dlt()
FUNCTION Void mbMenu()
FUNCTION Void mbSignup()
FUNCTION Boolean testNum(char [], int)
FUNCTION Void mbLogin(int&)
```

```
FUNCTION Void mbLoginMenu(int&)
FUNCTION Void mbbuyticket(int, int, double&)
FUNCTION Void print_seating_chart()
FUNCTION Double calculate_total_cost(int, int)
FUNCTION Boolean reserve_seat(int, int)
FUNCTION Void mbwallet(double&)
FUNCTION Void mbcancelacc(int, bool&)

MAIN FUNCTION()
        CALL FUNCTION admb()
        READ input stream
        RETURN 0

VOID FUNCTION admb()
BEGIN
        DECLARE Integer choiceA, num
        DECLARE Boolean choice INITIALIZE to false
        DECLARE Character quitChoice

        DO
                CHANGE console color to 0d
                DISPLAY “YSLL Cinema”
                DISPLAY “1. Admin Mode”
                DISPLAY “2. Member Mode”
                DISPLAY “3. Quit”
                DISPLAY “Please input a selection: ”
                READ choiceA

                IF choiceA == 1 THEN
                        DO
                                CLEAR console screen
                                CALL FUNCTION admMenu with num
                                IF num == 1 THEN
                                        CALL FUNCTION read()
```

```
                    ELSE IF num == 2 THEN
                        CALL FUNCTION add()
                    ELSE IF num == 3 THEN
                        CALL FUNCTION dlt()
                    ELSE IF num > 4 OR num < 0 THEN
                        DISPLAY “Invalid input. Please press ENTER to select
again!”
                        READ input stream
                        READ input stream
                    ELSE IF num ==  4 THEN
                        INITIALIZE choice to false
                    END IF
                WHILE num > 0 AND num != 4
                END WHILE
                CLEAR console screen

            ELSE IF choiceA == 2 THEN
                CLEAR console screen
                CALL FUNCTION mbMenu()

            ELSE IF choiceA == 3 THEN
                DISPLAY “Confirm to QUIT the system?(Y/N): “
                READ quitChoice
                WHILE quitChoice != 'Y'          AND quitChoice != 'y' AND
                    quitChoice != 'n' AND quitChoice != 'N'
                    DISPLAY “Invalid input. Please re-type again! (Y-Yes, N-No): “
                    READ quitChoice
                END WHILE
                IF quitChoice = ‘Y’ OR quitChoice = ‘y’
                    INITIALIZE choice to true
                END IF
                CLEAR console screen
            ELSE
                DISPLAY “Invalid input. Please press ENTER to select again!”
```

```
                READ input stream
                READ input stream
                CLEAR console screen
            END IF
        WHILE NOT choice
        END WHILE
END

VOID FUNCTION mbMenu()
BEGIN
        DECLARE choiceB, a
        DECLARE choice INITIALIZE to false
        DO
            CHANGE console color 0b
            DISPLAY "New User or Exist?"
            DISPLAY "1. New Register"
            DISPLAY "2. User Login"
            DISPLAY "Please input a selection: "
            READ choiceB

            IF choiceB == 1 THEN
                CLEAR console screen
                CALL FUNCTION mbSignup()
                CLEAR console screen
                INITIALIZE choice to true

            ELSE IF choiceB == 2 THEN
                CLEAR console screen
                CALL FUNCTION mbLogin with a
                CALL FUNCTION mbLoginMenu with a
                INITIALIZE choice to true

            ELSE
                DISPLAY "Invalid input. Please press ENTER to select again!"
```

```
                READ input stream
                READ input stream
                CLEAR console screen
            END IF
        WHILE NOT choice
        END WHILE
END

VOID FUNCTION mbSignup()
BEGIN
        DECLARE Integer i INITIALIZE to 0
        DECLARE Boolean valid INITIALIZE to false
        DECLARE memberSign member[arraySize]
        OPENFILE Outfile “member.txt” for WRITE and APPEND
        OPENFILE Infile “member.txt” for READ and APPEND

        WHILE NOT EOF Infile
            READFILE Infile, member[i].name
            READFILE Infile, member[i].password, pwSize
            i++
        END WHILE
        CLOSEFILE Infile

        CHANGE console color to 0b
        DISPLAY “Welcome To YSLL Cinema!”
        DISPLAY “New Member Sign Up”
        DISPLAY “Please ensure that you set the PASSWORD should be memorable!” SET red
color
        DISPLAY a new line

        DECLARE Boolean usernamevalid to false
        DECLARE Integer a

        DISPLAY “Username: ”
```

```
READ IGNORE newline character
READ member[i].name

DO
        FOR every a in i
                IF member[i].name != member[a].name
                        DISPLAY member[i].name
                        INITIALIZE usernamevalid to true

        FOR every a in I
                IF member[i].name == member[a].name
                        DISPLAY “This username has been created by other user.”
                        DISPLAY “Please re-type other username!”
                        DISPLAY new line
                        DISPLAY “Username: ”
                        READ member[i].name
                        READ IGNORE newline character
                        INITIALIZE usernamevalid to false

WHILE NOT usernamevalid

WRITEFILE Outfile, new line
WRITEFILE Outfile, member[i].name, end line
DISPLAY a new line

DO
        DISPLAY “L-Letter; N-Number” SET blue color
        DISPLAY “(Format - LLLLNNNN)” SET blue color
        DISPLAY “Password: ”
        READ member[i].password, pwSize

        IF CALL FUNCTION testNum with member[i].password, pwSize THEN
                WRITEFILE Outfile, member[i].password
                DISPLAY a new line
```

```
                    DISPLAY “Member successfully created! Your member ID is “
                    DISPLAY “Press ENTER to return back to menu: “
                    READ input stream
                    INITIALIZE valid to true

                ELSE
                    DISPLAY a new line
                    DISPLAY “Format incorrect! Please set the PASSWORD again.”

                END IF
        WHILE NOT valid
        END WHILE
        CLOSEFILE Outfile
END

BOOLEAN FUNCTION testNum with CHAR memberpw[], INT pwSize
BEGIN
        DECLARE Integer count
        FOR the first 4 counts in the password
                IF  each count of memberpw NOT alphabet
                        RETURN false
        END FOR

        FOR the last 4 counts in the password
                IF each count of memberpw NOT digit
                        RETURN false
        END FOR

        RETURN true
END

VOID FUNCTION mbLogin RETURNING a
BEGIN
        INITIALIZE a to 0
```

```
DECLARE Integer i to 0,  lgno to 0
DECLARE Boolean valid to false
DECLARE memberSign member[arraySize], login[arraySize]
OPENFILE Infile “member.txt” for READ and APPEND

WHILE NOT EOF Infile
        READFILE Infile, member[i].name
        READFILE Infile, member[i].password, pwSize
        i++
END WHILE
CLOSEFILE Infile

DO
        CHANGE console color to 0b
        DISPLAY “Welcome To YSLL Cinema!”
        DISPLAY “Member Sign In”
        DISPLAY a new line
        READ IGNORE newline character
        READ loginmb[lgno].name
        DISPLAY a new line
        DISPLAY “Password: ”
        READ loginmb[lgno].password, pwSize

        FOR every count ‘a’ read in member name and password
                IF member[a].name == loginmb[lgno].name
                AND member[a].password COMPARED with loginmb[lgno].password == 0
                        DISPLAY “You’re successfully log in! Press ENTER to MEMBER
MENU.”
                        READ input stream
                        CLEAR console screen
                        INITIALIZE valid to true
                        BREAK FOR LOOP
                END IF
```

```
                IF member[a].name != loginmb[lgno].name
                OR member[a].password COMPARED with loginmb[lgno].password == -1
                        DISPLAY “Not existing account, press ENTER to sign in again.”
                        READ input stream
                        CLEAR console screen
                END IF
        END FOR
    WHILE NOT valid
END

VOID FUNCTION mbLoginMenu RETURNING a
BEGIN
    DECLARE Integer choiceC, current_idno to a
    DECLARE Double balance to 0
    DECLARE movieType movie[arraySize]
    DECLARE Boolean loginChoice to false, auto_logout to false, buyticketChoice to false

    DO
        CHANGE console color to 0b
        DISPLAY “YSLL Cinema - Member Login Menu”
        DISPLAY “1. Movie List”
        DISPLAY “2. Buy Ticket”
        DISPLAY “3. Wallet”
        DISPLAY “4. Delete Acoount”
        DISPLAY “5. Logout”
        DISPLAY “Please input a selection: ”
        READ choiceC
        DISPLAY new line

        IF choiceC == 1 THEN
                CALL FUNCTION read()

        ELSE IF choiceC == 2 THEN
                DO
```

```
        CLEAR console screen
        OPENFILE Infile “cinema.txt” for READ and APPEND
        DECLARE Integer i to 0, movieNo

        DISPLAY “1. Buy Ticket”
        DISPLAY “Movie List”
        DISPLAY “Name Price”
        IF Infile is OPEN
                WHILE NOT EOF Infile
                        READFILE Infile, movie[i].name
                        READFILE Infile, movie[i].price
                        DISPLAY i+1, “.”, movie[i].name
                        DISPLAY FIXED, SETPRECISION to 2
                        DISPLAY “RM”, movie[i].price
                        i++
                        READFILE IGNORE newline character
                END WHILE
        END IF
        CLOSEFILE Infile

        DISPLAY “Choose a movie: ”
        READ movieNo

        IF movieNo <= I THEN
                CALL FUNCTION mbbuyticket with movieNo, i, balance
                CLEAR console screen
                INITIALIZE buyticketChoice to true

        ELSE
                DISPLAY “Invalid input. Press ENTER to re-select again!”
                READ input stream
                READ input stream (let user press ENTER)
        END IF
WHILE NOT buyticketChoice
```

```
                END WHILE

            ELSE IF choiceC == 3 THEN
                CALL FUNCTION mbwallet with balance
                CLEAR console screen

            ELSE IF choiceC == 4 THEN
                CALL FUNCTION mbcancelacc with current_idno, auto_logout
                CLEAR console screen
                IF auto_logout is true
                    INITIALIZE loginChoice to true
                END IF

            ELSE IF choiceC == 5 THEN
                DISPLAY "Thank for your support to our YSLL Cinema!"
                DISPLAY "Press ENTER to proceed logout."
                READ input stream
                READ input stream (let user press ENTER)
                CLEAR console screen
                INITIALIZE loginChoice to true
            END IF
        WHILE NOT loginChoice
        END WHILE
END

VOID FUNCTION mbbuyticket with Integer movieNo, i RETURNING Double balance
BEGIN
        DECLARE movieType movie[arraySize]
        CLEAR console screen
        CHANGE console color to 0d

        DECLARE Integer date, time, numSeats = 0
        DECLARE Double payment, totalPayment, change
        DECLARE Character choice
```

```
WHILE true
        DISPLAY “Date Available”
        DISPLAY “1. 16/4/2023”
        DISPLAY “2. 17/4/2023”
        DISPLAY “3. 18/4/2023”
        DISPLAY “4. 19/4/2023”
        DISPLAY “5. 20/4/2023”
        DISPLAY “Select a date: ”
        READ date

        IF date >= 1 AND date <= 5 THEN
                BREAK WHILE
        ELSE
                DISPLAY “Please enter a number between 1 to 5 only”
        END IF
END WHILE
DISPLAY new line
CLEAR console screen

WHILE true
        DISPLAY “You have selected the date”, date, “.”
        SWITCH date
        CASE 1
                DISPLAY “16/4/2023”
                BREAK SWITCH
        CASE 2
                DISPLAY “17/4/2023”
                BREAK SWITCH
        CASE 3
                DISPLAY “18/4/2023”
                BREAK SWITCH
        CASE 4
                DISPLAY “19/4/2023”
```

```
        BREAK SWITCH
CASE 5
        DISPLAY “20/4/2023”
        BREAK SWITCH
END CASE

DISPLAY new line
DISPLAY “Do you want to change your selection? (Y/N): ”
DECLARE Character choice
READ choice
CLEAR console screen

IF choice == ‘Y’ OR choice == ‘y’ THEN
        WHILE true
                DISPLAY “Date Available”
                DISPLAY “1. 16/4/2023”
                DISPLAY “2. 17/4/2023”
                DISPLAY “3. 18/4/2023”
                DISPLAY “4. 19/4/2023”
                DISPLAY “5. 20/4/2023”
                DISPLAY “Select a date: ”
                READ date

                IF date >= 1 && date <= 5 THEN
                        BREAK WHILE LOOP
                ELSE
                        DISPLAY “Please enter a number between 1 to 5 only”
        END IF
        DISPLAY new line
        CLEAR console screen
        END WHILE

ELSE IF         choice == ‘N’ OR choice == ‘n’ THEN
        BREAK WHILE
```

```
        ELSE
                DISPLAY “Invalid input. Please enter Y or N only.”
END IF
END WHILE

DISPLAY “You have selected date”, date, “.”
SWITCH date
CASE 1
        DISPLAY “16/4/2023”
        BREAK SWITCH
CASE 2
        DISPLAY “17/4/2023”
        BREAK SWITCH
CASE 3
        DISPLAY “18/4/2023”
        BREAK SWITCH
CASE 4
        DISPLAY “19/4/2023”
        BREAK SWITCH
CASE 5
        DISPLAY “20/4/2023”
        BREAK SWITCH
END CASE
DISPLAY new line
CLEAR console screen

CHANGE console color to 0e
WHILE true
        DISPLAY “Time available”
        DISPLAY “1. 10:00AM”
        DISPLAY “2. 01:15PM”
        DISPLAY “3. 01:45PM”
        DISPLAY “4. 03:45PM”
```

```
        DISPLAY “5. 04:15PM”
        DISPLAY “6. 06:45PM”
        DISPLAY “7. 09:15PM”
        DISPLAY “Select a time: ”
        READ time

        IF time >= 1 AND time <= 7 THEN
                BREAK WHILE
        ELSE
                DISPLAY “Please enter a number between 1 to 7 only”
END IF
DISPLAY new line
CLEAR console screen
END WHILE

WHILE true
        DISPLAY “You have selected the time”, time, “.”
        SWITCH time
        CASE 1
                DISPLAY “10:00AM”
                BREAK SWITCH
        CASE 2
                DISPLAY “01:15PM”
                BREAK SWITCH
        CASE 3
                DISPLAY “01:45PM”
                BREAK SWITCH
        CASE 4
                DISPLAY “03:45PM”
                BREAK SWITCH
        CASE 5
                DISPLAY “04:15PM”
                BREAK SWITCH
        CASE 6:
```

```
        DISPLAY “06:45PM”
        BREAK SWITCH
CASE 7
        DISPLAY “09:15PM”
        BREAK SWITCH
END CASE

DISPLAY new line
DISPLAY “Do you want to change your selection? (Y/N): ”
DECLARE Character choice
READ choice
CLEAR console screen

IF choice == ‘Y’ OR choice == ‘y’ THEN
        WHILE true
                DISPLAY “Time available”
                DISPLAY “1. 10:00AM”
                DISPLAY “2. 01:15PM”
                DISPLAY “3. 01:45PM”
                DISPLAY “4. 03:45PM”
                DISPLAY “5. 04:15PM”
                DISPLAY “6. 06:45PM”
                DISPLAY “7. 09:15PM”
                DISPLAY “Select a time: ”
                READ time

                IF time >= 1 AND time <= 7 THEN
                        BREAK WHILE
                ELSE
                        DISPLAY “Please enter a number between 1 to 7 only”
                END IF
        END WHILE
DISPLAY new line
CLEAR console screen
```

```
        ELSE IF choice == ‘N’ OR choice == ‘n’ THEN
                BREAK WHILE

        ELSE
                DISPLAY “Invalid input. Please enter Y or N only.”
        END IF
END WHILE

DISPLAY “You have selected time”, time, “.”
SWITCH time
CASE 1
        DISPLAY “10:00AM”
        BREAK SWITCH
CASE 2
        DISPLAY “01:15PM”
        BREAK SWITCH
CASE 3
        DISPLAY “01:45PM”
        BREAK SWITCH
CASE 4
        DISPLAY “03:45PM”
        BREAK SWITCH
CASE 5
        DISPLAY “04:15PM”
        BREAK SWITCH
CASE 6:
        DISPLAY “06:45PM”
        BREAK SWITCH
CASE 7
        DISPLAY “09:15PM”
        BREAK SWITCH
END CASE
DISPLAY new line
```

```
CLEAR console screen
DISPLAY new line

DECLARE Character choice to ‘y’
DO
        CHANGE console color to 0e
        DISPLAY “SCREEN”

        CALL FUNCTION print_seating_chart()
        DISPLAY new line
        DISPLAY “Enter column number (1-”, columns, “): ”
        DECLARE Integer seat
        READ seat
        DISPLAY “Enter row number (1-”, rows, “): ”
        DECLARE Integer row
        READ row

        IF row < 1 OR row > rows OR seat < 1 OR seat > columns THEN
                DISPLAY “Invalid input. Press ENTER to select again.”
                READ input stream
                READ input stream (let user press ENTER)
                CLEAR console screen
                INITIALIZE choice to ‘y’
                CONTINUE WHILE
        END IF

        DECLARE Boolean reserved to CALL FUNCTION reserve_seat with row, seat
        IF reserved is true THEN
                DISPLAY “Seat reserved.”
                numSeats++
        ELSE
                DISPLAY “Seat is already reserved. Press ENTER to select again!”
                DISPLAY new line
                READ input stream
```

```
                READ input stream (let user press ENTER)
                CLEAR console screen
                CONTINUE WHILE
        END IF

        DISPLAY “Do you want to reserve more seats? (Y/N): ”
        READ choice
        CLEAR console screen
    WHILE choice == ‘Y’ OR choie == ‘y’
    END WHILE

    INITIALIZE totalPayment to CALL FUNCTION calculate_total_cost with numSeats,
movieN

    DISPLAY newline, newline
    DISPLAY “Total price is: RM”, totalPayment
    DISPLAY “Wallet balance is: RM”, balance

    CALCULATE change = balance - totalPayment
    DECLARE Boolean payment_successful to true

    IF change >= 0 THEN
        INITIALIZE balance to change
        DISPLAY “Change: RM”, change
        DISPLAY “Thank you for purchasing and have a good day :)!”
        DISPLAY new line

    ELSE
        DISPLAY “Error: Insufficient amount.”
        DISPLAY “Payment Unsuccessful, please top up WALLET before transaction”
        DISPLAY “Press ENTER return back to Member Menu.”
        READ input stream
        READ input stream (let user press ENTER)
        INITIALIZE payment_successful to false
```

```
END IF
DISPLAY new line

WHILE payment_successful is true
        DISPLAY “YSLL Cinema Receipt”
        DISPLAY “Movie Title: ”
        OPENFILE Infile “cinema.txt” for READ
        WHILE NOT EOF Infile
                FOR every a in movie list
                        READFILE Infile, movie[a].name
                        READFILE Infile, movie[a].price
                        READFILE IGNORE newline character
                END FOR
        END WHILE
        CLOSEFILE Infile

        DISPLAY movie[movieNo - 1].name
        DISPLAY new line

        DISPLAY “Movie Date: ”
        SWITCH date
        CASE 1
                DISPLAY “16/4/2023”
                BREAK SWITCH
        CASE 2
                DISPLAY “17/4/2023”
                BREAK SWITCH
        CASE 3
                DISPLAY “18/4/2023”
                BREAK SWITCH
        CASE 4
                DISPLAY “19/4/2023”
                BREAK SWITCH
        CASE 5
```

```
        DISPLAY “20/4/2023”
        BREAK SWITCH
END CASE
DISPLAY new line

DISPLAY “Movie Time: ”
SWITCH time
CASE 1
        DISPLAY “10:00AM”
        BREAK SWITCH
CASE 2
        DISPLAY “01:15PM”
        BREAK SWITCH
CASE 3
        DISPLAY “01:45PM”
        BREAK SWITCH
CASE 4
        DISPLAY “03:45PM”
        BREAK SWITCH
CASE 5
        DISPLAY “04:15PM”
        BREAK SWITCH
CASE 6:
        DISPLAY “06:45PM”
        BREAK SWITCH
CASE 7
        DISPLAY “09:15PM”
        BREAK SWITCH
END CASE

DISPLAY new line

DISPLAY “Total payment: RM”, totalPayment
DISPLAY “Show this ticket to the counter to clarify”
```

```
            DISPLAY “the ticket validity due to security reason”
            DISPLAY “then only assign according to the seating”
            DISPLAY “position. Sorry for problem caused.”
            DISPLAY “Have a great day! :) QR CODE”

            READ input stream
            READ input stream (let user press any key)
            INITIALIZE payment_successful to false
        END WHILE
END

VOID FUNCTION print_seating_chart()
BEGIN
        DISPLAY new line
        FOR every Integer i in rows
            FOR every Integer j in columns
                IF seats[i][j] == 0 THEN
                        DISPLAY “0”
                ELSE
                        DISPLAY “X”
                END IF
            END FOR
        END FOR
        DISPLAY new line
END

DOUBLE FUNCTION calculate_total_cost with Integer numSeats, movieNo
BEGIN
        DECLARE movieType movie[arraySize]
        OPENFILE Infile “cinema.txt” for READ
        DECLARE Integer i = 0
        WHILE NOT EOF Infile
            READFILE IGNORE newline character
            READFILE Infile, movie[i].name
```

```
            READFILE Infile, movie[i].price
            i++
        END WHILE
        RETURN numSeats * movie[movieNo - 1].price
END

BOOLEAN FUNCTION reserve_seat with Integer row, seat
BEGIN
        IF seats[row-1][seat-1] == 0 THEN
            seats[row-1][seat-1] = 1
            RETURN true
        ELSE
            RETURN false
        END IF
END

VOID FUNCTION mbwallet RETURNING balance
BEGIN
        DECLARE Integer topup
        DECLARE Character topupChoice
        DECLARE Boolean topupexit to false

        CLEAR console screen
        DISPLAY “Wallet”
        DISPLAY FIXED, SETPRECISION to 2
        DISPLAY “Blance = RM”, balance
        DISPLAY “Would you like to top up your wallet? (Y/N): “
        READ topupChoice
        WHILE topupChoice != ‘Y’ AND topupChoice != ‘y’
        AND topupChoice != ‘N’ AND topupChoice != ‘n’
            DISPLAY “Invalid input. Please re-type again! (Y-Yes, N-No): “
            READ topupChoice
        END WHILE
```

```
IF topupChoice == 'Y' OR topupChoice == 'y' THEN
        CLEAR console screen
        DISPLAY "Top Up Service"
        DISPLAY "RM10"
        DISPLAY "RM20"
        DISPLAY "RM50"
        DISPLAY "RM100"

        DO
                DISPLAY new line
                DISPLAY "Please key in a value to TOPUP: "
                READ topup

                SWITCH topup
                CASE 10
                        CALCULATE balance += topup
                        BREAK SWITCH
                CASE 20
                        CALCULATE balance += topup
                        BREAK SWITCH
                CASE 50
                        CALCULATE balance += topup
                        BREAK SWITCH
                CASE 100
                        CALCULATE balance += topup
                        BREAK SWITCH
                DEFAULT
                        DISPLAY "Invalid input. Please re-type TOPUP value!"
                        CONTINUE WHILE
                END CASE

                DISPLAY "Would you like to TOP UP again? (Y/N): "
                READ topupChoice
```

```
                    WHILE topupChoice != 'Y' AND topupChoice != 'y'
                    AND topupChoice != 'N' AND topupChoice != 'n'
                        DISPLAY "Invalid input. Please re-type again! (Y-Yes, N-No): "
                        READ topupChoice
                    END WHILE

                    IF topupChoice == 'N' OR topupChoice == 'n'
                        DISPLAY "Press ENTER to return back to Member MENU."
                        READ input stream
                        READ input stream (let user to press ENTER)
                        INITIALIZE topupexit to true
                    END IF
                WHILE NOT topupexit
                END WHILE

        ELSE IF topupChoice == 'N' OR topupChoice == 'n' THEN
                DISPLAY "Press ENTER to return back to Member MENU."
                READ input stream
                READ input stream (let user to press ENTER)
        END IF
END

VOID FUNCTION mbcancelacc with Integer current_idno RETURN Boolean auto_logout
BEGIN
        DECLARE Integer i = 0
        DECLARE Character deleteChoice
        DECLARE memberSign member[arraySize]
        OPENFILE Infile "member.txt" for READ and APPEND

        WHILE NOT EOF Infile
                READFILE Infile, member[i].name
                READFILE Infile, member[i].password, pwSize
                i++
        END WHILE
```

```
CLOSEFILE Infile

DISPLAY “Do you want to delete your current account? (Y/N): ”
READ deleteChoice

WHILE deleteChoice != ‘Y’ AND deleteChoice != ‘y’
AND deleteChoice != ‘N’ AND deleteChoice != ‘n’
        DISPLAY “Invalid input. Please re-type again! (Y-Yes, N-No): “
        READ deleteChoice
END WHILE

IF deleteChoice == ‘Y’ OR deleteChoiec == ‘y’
        OPENFILE Outfile “member.txt” for TRUNCATE
        FOR every Integer a in i-1 (total member)
                IF member[a].name == member[current_idno].name AND
                member[a].password COMPARED member[current_idno].password == 0
                        INITIALIZE member[a].name to member[a+1].name
                        member[a].password COPY value to member[a+1].password

                        IF a != 0 THEN
                                WRITEFILE Outfile, new line
                        END IF
                        WRITEFILE Outfile, member[a].name
                        WRITEFILE Outfile, member[a].password

                ELSE IF member[a].name != member[current_idno].name OR
                member[a].password COMPARED member[current_idno].password == -1
                        IF a != 0 THEN
                                WRITEFILE Outfile, new line
                        END IF
                        WRITEFILE Outfile, member[a].name
                        WRITEFILE Outfile, member[a].password
                END IF
        END FOR
```

```
            CLOSEFILE Outfile
            DISPLAY “Account deleted. Press ENTER auto logout…”
            READ input stream
            READ input stream (let user to press ENTER)
            INITIALIZE auto_logout to true

        ELSE IF deleteChoice == ‘N’ OR deleteChoice == ‘n’
            DISPLAY “Press ENTER return back to Member MENU.”
            READ input stream
            READ input stream (let user to press ENTER)

        END IF
END

VOID FUNCTION admMenu RETURN Integer choice
BEGIN
        CHANGE console color to 0a
        DISPLAY “YSLL Cinema - Admin Operating Menu”
        DISPLAY “1. Movie List”
        DISPLAY “2. Add Movie”
        DISPLAY “3. Delete Movie”
        DISPLAY “4. Back”
        DISPLAY “Please input a selection: ”
        READ choice
END

VOID FUNCTION read()
BEGIN
        OPENFILE Infile “cinema.txt” for READ and APPEND
        DECLARE movieType movie[arraySize]
        DECLARE Integer i to 0

        CLEAR console screen
        DISPLAY “Movie List”
```

```
        DISPLAY “Name Price”

        IF Infile is open THEN
                WHILE NOT EOF Infile
                        READFILE Infile, movie[i].name
                        READFILE Infile, movie[i].price
                        DISPLAY i+1, “.”, movie[i].name
                        DISPLAY FIXED, SETPRECISION to 2
                        DISPLAY “RM”, movie[i].price
                        i++
                        READFILE IGNORE newline character
                END WHILE
                CLOSEFILE Infile

        ELSE
                DISPLAY “Unable to open storing file!”
        END IF

        DISPLAY “Press ENTER to return back to Operating Menu.”
        READ input stream
        READ input stream (let user to press ENTER)
        CLEAR console screen
END

VOID FUNCTION add()
BEGIN
        OPENFILE Infile “cinema.txt” for READ and APPEND
        OPENFILE Outfile “cinema.txt” for WRITE and APPEND
        DECLARE movieType movie[arraySize]
        DECLARE Character addMovie

        DO
                DECLARE Integer i to 0
                WHILE NOT EOF Infile
```

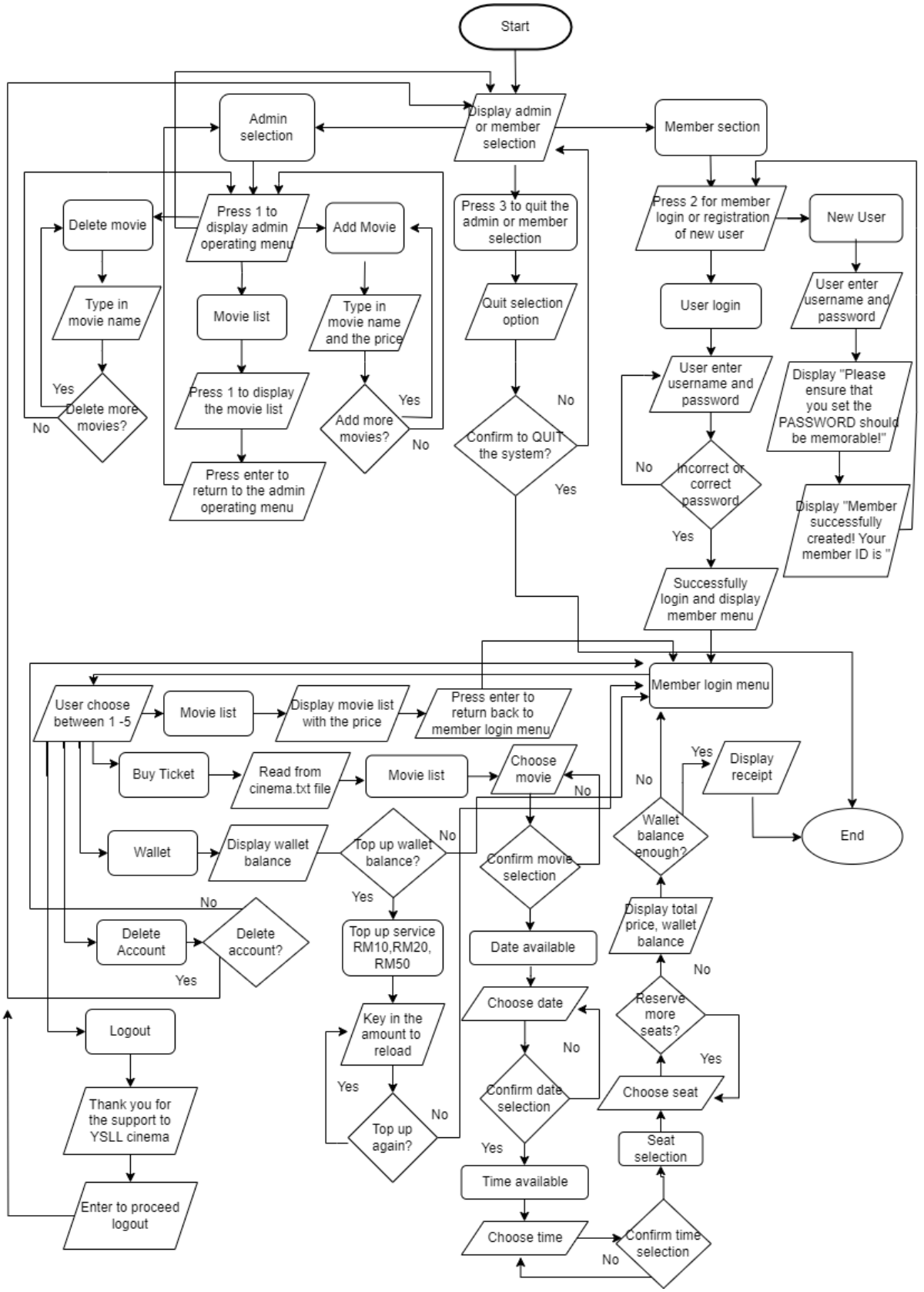
```
            READFILE Infile, movie[i].name
            READFILE Infile, movie[i].price
            i++
            READFILE IGNORE newline character
        END WHILE

        DISPLAY new line
        DISPLAY “Add Movie”
        DISPLAY “Name: ”
        WRITEFILE new line
        READ IGNORE newline character
        READ movie[i].name
        WRITEFILE Outfile, movie[i].name
        DISPLAY “Price: ”
        READ movie[i].price
        WRITEFILE Outfile, movie[i].price

        DISPLAY “Do you want to add more movie?(Y/N): ”
        READ addMovie

        WHILE addMovie != ‘Y’ AND addMovie != ‘y’
        AND addMovie != ‘N’ AND addMovie != ‘n’
            DISPLAY “Invalid input. Please type again (Y-Yes, N-No): ”
            READ addMovie
        END WHILE

WHILE addMovie == ‘Y’ OR addMovie ==’y’
END WHILE
CLOSEFILE Infile
CLOSEFILE Outfile

DISPLAY new line
DISPLAY “Press ENTER to return back to Operating Menu.”
READ input stream
```

```
        READ input stream (let user to press ENTER)
        CLEAR console screen
END

VOID FUNCTION dlt()
BEGIN
        DECLARE movieType movie[arraySize]
        DECLARE Integer i, a to 0, dlno
        DECLARE Boolean choice to false

        DO
                OPENFILE Infile "cinema.txt" for READ and APPEND
                INITIALIZE i to 0
                CLEAR console screen
                DISPLAY "Please choose a movie to delete."
                DISPLAY "Movie List"
                DISPLAY "Name Price"

                IF Infile is open THEN
                        WHILE NOT EOF Infile
                                READFILE Infile, movie[i].name
                                READFILE Infile, movie[i].price
                                DISPLAY i+1, ".", movie[i].name
                                DISPLAY FIXED, SETPRECISION to 2
                                DISPLAY "RM", movie[i].price
                                i++
                                READFILE IGNORE newline character
                        END WHILE
                        DISPLAY new line
                        CLOSEFILE Infile
                ELSE
                        DISPLAY "Unable to open storing file!"
                END IF
```

```
        DISPLAY “Which one you want to delete? (Type MovieNo to Delete) : ”
        READ dlno

        IF dlno <= I AND dlno >0 THEN
                INITIALIZE choice to true

        ELSE IF dlno > i OR dlno <= 0 THEN
                DISPLAY “Invalid input. Press ENTER to select again!”
                READ input stream
                READ input stream (let user to press ENTER)
        END IF
WHILE NOT choice
END WHILE

CALCULATE dlno -= 1

OPENFILE Outfile “cinema.txt” for TRUNCATE

WHILE a < i - 1
        DECLARE Integer pos to find movie[dlno].name exist inside movie[a].name
        IF pos found THEN
                WHILE a < i - 1
                        REPLACE movie[a].name to movie[a+1].name
                        REPLACE movie[a].price to movie[a+1].price

                        IF a != 0
                                WRITEFILE Outfile, new line
                        WRITEFILE Outfile, movie[a].name
                        WRITEFILE Outfile, movie[a].price
                        INITIALIZE pos to not find
                        a++
                END WHILE

        ELSE IF pos not found AND a != i -1 THEN
```

```
                IF a != 0
                        WRITEFILE Outfile, new line
                WRITEFILE Outfile, movie[a].name
                WRITEFILE Outfile, movie[a].price
                a++
            END IF
        END WHILE

        CLOSEFILE Outfile
        DISPLAY “Press ENTER to return back to Operating Menu.”
        READ input stream
        READ input stream (let user to press ENTER)
        CLEAR console screen
END
```

v. Flowchart


Start
Display admin or member selection
Admin selection
Member section
Press 1 to display admin operating menu
Delete movie
Add Movie
Type in movie name
Movie list
Type in movie name and the price
Yes
Delete more movies?
No
Press 1 to display the movie list
Add more movies?
Yes
No
Press enter to return to the admin operating menu
Press 3 to quit the admin or member selection
Quit selection option
Confirm to QUIT the system?
No
Yes
Press 2 for member login or registration of new user
New User
User login
User enter username and password
User enter username and password
Display "Please ensure that you set the PASSWORD should be memorable!"
No
Incorrect or correct password
Yes
Display "Member successfully created! Your member ID is "
Successfully login and display member menu
Member login menu
User choose between 1 -5
Movie list
Display movie list with the price
Press enter to return back to member login menu
Buy Ticket
Read from cinema.txt file
Movie list
Choose movie
No
Yes
Display receipt
End
Wallet
Display wallet balance
Top up wallet balance?
No
Yes
Confirm movie selection
Wallet balance enough?
No
Yes
Delete Account
Delete account?
No
Yes
Top up service RM10,RM20, RM50
Date available
Display total price, wallet balance
Logout
Key in the amount to reload
Choose date
Reserve more seats?
No
Yes
Thank you for the support to YSLL cinema
Yes
Top up again?
No
Confirm date selection
No
Choose seat
Enter to proceed logout
Yes
Time available
Seat selection
Choose time
Confirm time selection
No

vi. Test Case

1. Add movie

```
|-----------------------------------------------------|
|                     Movie List                      |
|-----------------------------------------------------|
| Name                                        | Price |
|-----------------------------------------------------|
| 1. Spiderman: No Way Home
|                                             | RM21.40 |
| 2. Demon Slayer: Kimetsu No Yaiba
|                                             | RM23.00 |
| 3. John Wick: Chapter 4
|                                             | RM18.70 |
| 4. The Conjuring: Chapter 3
|                                             | RM17.00 |
| 5. Love Yourself: HYNH
|                                             | RM24.00 |
| 6. Interstellar
|                                             | RM20.70 |
|-----------------------------------------------------|
Press ENTER to return back to Operating Menu.
```

Original

```
    |---------------------------------------|
    |  YSLL Cinema - Admin Operating Menu   |
    |---------------------------------------|
    |1. Movie List                          |
    |2. Add Movie                           |
    |3. Delete Movie                        |
    |4. Back                                |
    |---------------------------------------|
        Please input a selection: 2

    Add Movie
    Name: The Boy in the Striped Pajamas
    Price: 20.8
    Do you want to add more movie?(Y/N): y

    Add Movie
    Name: Tenet
    Price: 20.2
    Do you want to add more movie?(Y/N): n

    Press ENTER to return back to Operating Menu.
```

Input

```
|-----------------------------------------------------|
|                     Movie List                      |
|-----------------------------------------------------|
| Name                                        | Price |
|-----------------------------------------------------|
| 1. Spiderman: No Way Home
|                                             | RM21.40 |
| 2. Demon Slayer: Kimetsu No Yaiba
|                                             | RM23.00 |
| 3. John Wick: Chapter 4
|                                             | RM18.70 |
| 4. The Conjuring: Chapter 3
|                                             | RM17.00 |
| 5. Love Yourself: HYNH
|                                             | RM24.00 |
| 6. Interstellar
|                                             | RM20.70 |
| 7. The Boy in the Striped Pajamas
|                                             | RM20.80 |
| 8. Tenet
|                                             | RM20.20 |
|-----------------------------------------------------|
Press ENTER to return back to Operating Menu.
```

Output

2. Delete movie

```
|-----------------------------------------------------|
|                      Movie List                     |
|-----------------------------------------------------|
| Name                                    | Price     |
|-----------------------------------------------------|
| 1. Spiderman: No Way Home
|                                         | RM21.40 |
| 2. Demon Slayer: Kimetsu No Yaiba
|                                         | RM23.00 |
| 3. John Wick: Chapter 4
|                                         | RM18.70 |
| 4. The Conjuring: Chapter 3
|                                         | RM17.00 |
| 5. Love Yourself: HYNH
|                                         | RM24.00 |
| 6. Interstellar
|                                         | RM20.70 |
| 7. The Boy in the Striped Pajamas
|                                         | RM20.80 |
| 8. Tenet
|                                         | RM20.20 |
|-----------------------------------------------------|
Press ENTER to return back to Operating Menu.
```

Original

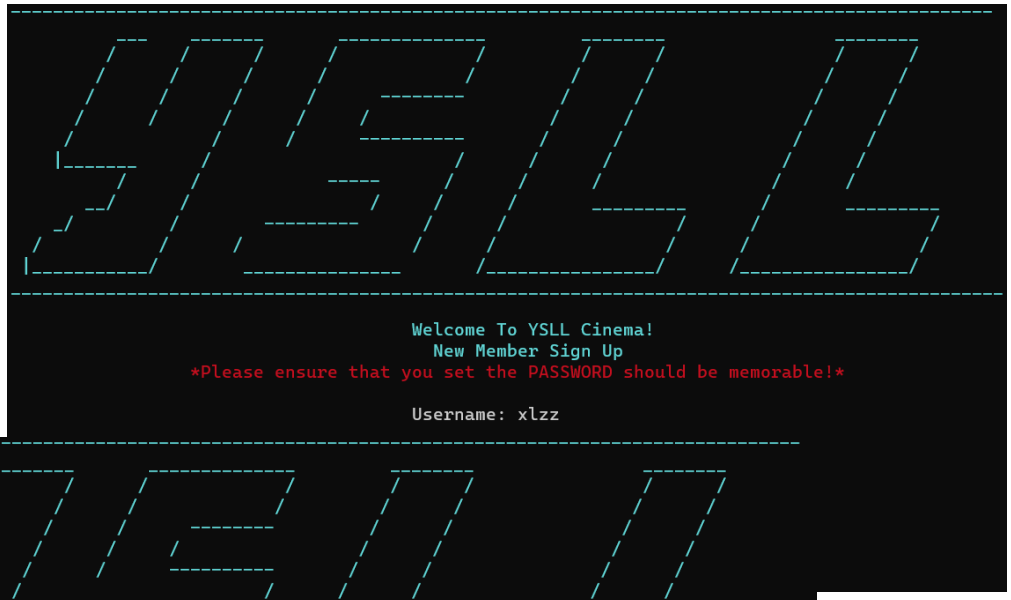
```
Please choose a movie to delete.
|-----------------------------------------------------|
|                      Movie List                     |
|-----------------------------------------------------|
| Name                                    | Price     |
|-----------------------------------------------------|
| 1. Spiderman: No Way Home
|                                         | RM21.40 |
| 2. Demon Slayer: Kimetsu No Yaiba
|                                         | RM23.00 |
| 3. John Wick: Chapter 4
|                                         | RM18.70 |
| 4. The Conjuring: Chapter 3
|                                         | RM17.00 |
| 5. Love Yourself: HYNH
|                                         | RM24.00 |
| 6. Interstellar
|                                         | RM20.70 |
| 7. The Boy in the Striped Pajamas
|                                         | RM20.80 |
| 8. Tenet
|                                         | RM20.20 |
|-----------------------------------------------------|

Which one you want to delete? (Type MovieNo to Delete) : 4
Press ENTER to return back to Operating Menu.
```

Input

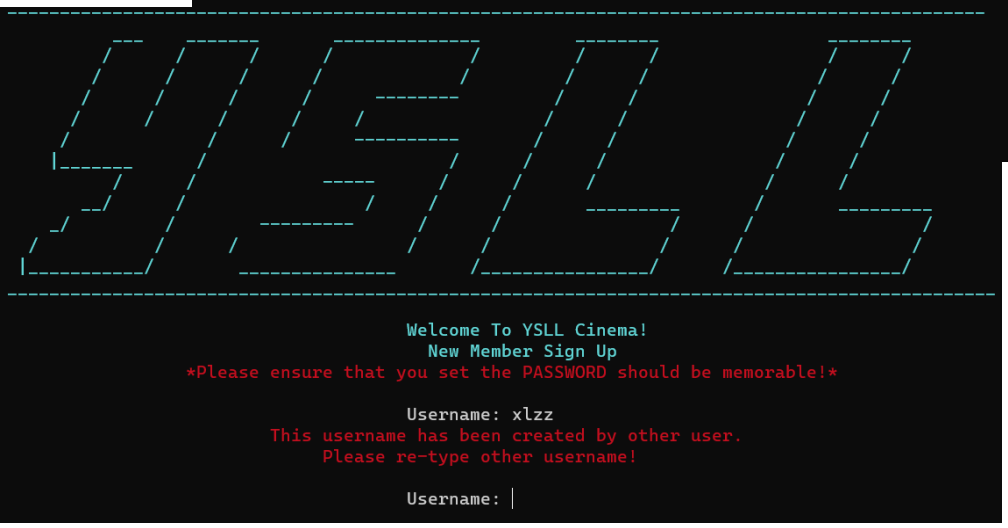
```
|-----------------------------------------------------|
|                      Movie List                     |
|-----------------------------------------------------|
| Name                                    | Price     |
|-----------------------------------------------------|
| 1. Spiderman: No Way Home
|                                         | RM21.40 |
| 2. Demon Slayer: Kimetsu No Yaiba
|                                         | RM23.00 |
| 3. John Wick: Chapter 4
|                                         | RM18.70 |
| 4. Love Yourself: HYNH
|                                         | RM24.00 |
| 5. Interstellar
|                                         | RM20.70 |
| 6. The Boy in the Striped Pajamas
|                                         | RM20.80 |
| 7. Tenet
|                                         | RM20.20 |
|-----------------------------------------------------|
Press ENTER to return back to Operating Menu.
```

Output

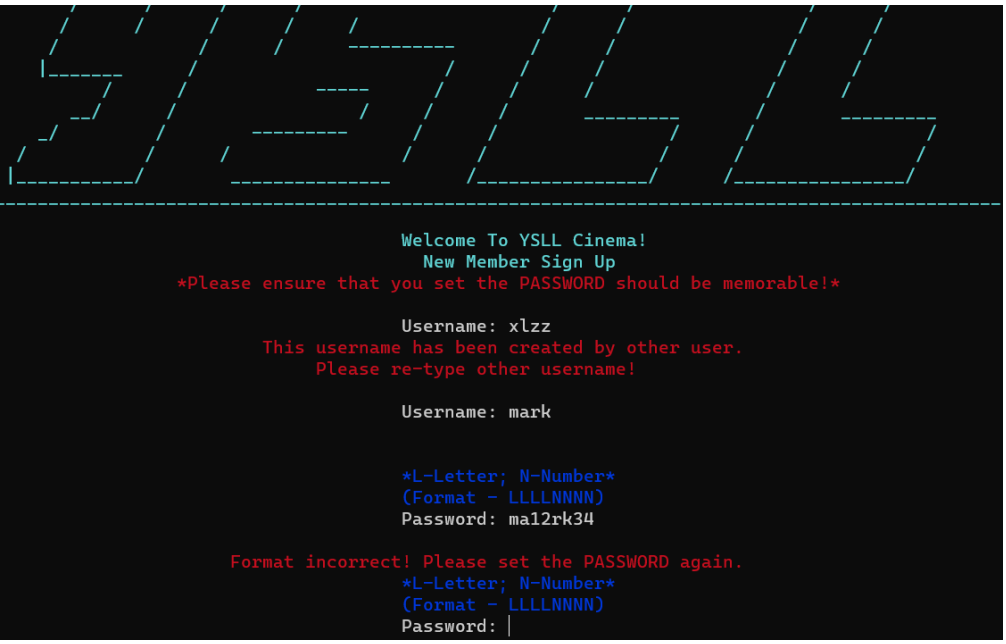
3. User register



Register Successful



Repeat Username (Invalid)



Password Format (Invalid)

4. User Login

Login Successful

Login Unsuccessful

5. Buy Ticket

```
Total price is: RM20.70
Wallet balance is: RM70.00
Change: RM49.30
Thank you for purchasing and have a good day :)!


--------------------------------------------------
*              YSLL Cinema Receipt               *
--------------------------------------------------
*Movie Title: Interstellar
*Movie Date: 16/4/2023
*Movie Time: 01:15 PM
|                                                |
|Total payment:RM20.70                           |
|                                                |
|Show this ticket to the counter to clarify      |
|the ticket validity due to security reason      |
|then only assign according to the seating       |
|postion. Sorry for problem caused.              |
|                                                |
|                                                |
|Have a great day! :)           QR CODE          |
|                               *********        |
|                               *       *        |
|                               *       *        |
|                               *********        |
|                                                |
--------------------------------------------------
```

Receipt

6. Top Up Wallet

```
Total price is: RM37.40
Wallet balance is: RM0.00
Error: Insufficient amount.
Payment Unsuccessful, please top up WALLET before transaction.
Press ENTER return back to Member Menu.
```

Insufficient amount to buy ticket

```
|----------------|
| Top Up Service |
|----------------|
|      RM10      |
|      RM20      |
|      RM50      |
|      RM100     |
|----------------|

Please key in a value to TOPUP: 20
Would you like to TOP UP again? (Y/N): y

Please key in a value to TOPUP: 50
Would you like to TOP UP again? (Y/N): n
Press ENTER to return back to Member MENU.
```

Choose Top Up Value

```
|------------------------------|
|            Wallet            |
|------------------------------|
      Balance = RM70.00
|------------------------------|
Would you like to top up your wallet?(Y/N):
```

Wallet after Top Up

7. Seat

```
----------------------------------------
|                                      |
----------------------------------------
                SCREEN


O O O O O O O O O O O O O O O O O O O O
X O O O O O O O O O O O O O O O O O O O
O O O O O O O O O O O O O O O O O O O O
O O O O O O O O O O O O O O O O O O O O
O O O O O O O O O O O O O O O O O O O O
O O O O O O O O O O O O O O O O O O O O
O O O O O O O O O O O O O O O O O O O O
O O O O O O O O O O O O O O O O O O O O
O O O O O O O O O O O O O O O O O O O O
O O O O O O O O O O O O O O O O O O O O

Enter column number (1-20): 1
Enter row number (1-10): 2
Seat is already reserved. Press ENTER to select again!
```

Seat unavailable.

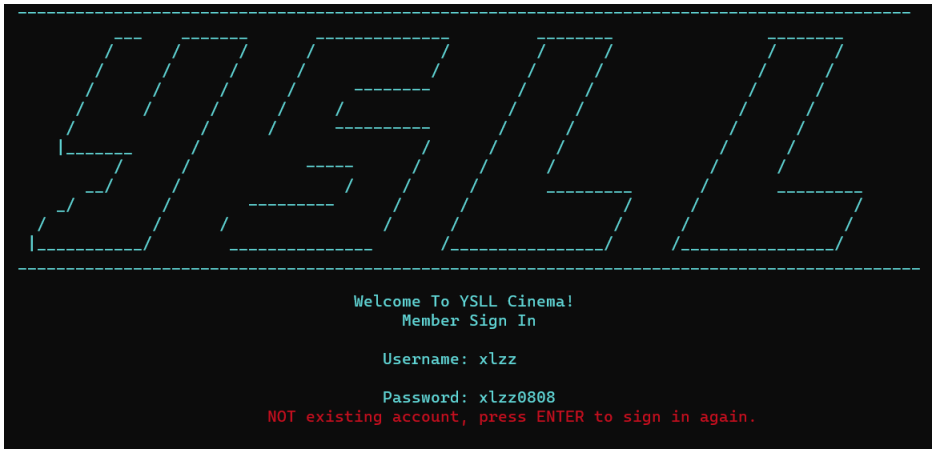
Can choose multiple seat.

8. Delete Account

```
    |--------------------------------------|
    |  YSLL Cinema - Member Login Menu     |
    |--------------------------------------|
    |1. Movie List                         |
    |2. Buy Ticket                         |
    |3. Wallet                             |
    |4. Delete Account                     |
    |5. Logout                             |
    |--------------------------------------|
    Please input a selection: 4

    Do you want to delete your current account? (Y/N): y
    Account deleted. Press ENTER auto logout...
```

Delete current account

Account deleted, can’t login anymore



vii. Source Code

```
#include<iostream>
#include<string>
#include<fstream>
#include<iomanip>
#include<cctype>
#include<cstring>
#define _CRT_SECURE_NO_WARNINGS //VS secure version to use strcpy function
using namespace std;

//global variable
const int arraySize = 1000;
const int pwSize = 9;

const int rows = 10;
const int columns = 20;
int seats[rows][columns] = { 0 };

struct movieType {
    string name;
    double price;
};

struct memberSign {
    string name;
    char password[pwSize];
};

//function prototype
void admb();
void admMenu(int&);
void read();
```

```
void add();
void dlt();
void mbMenu();
void mbSignup();
bool testNum(char [], int);
void mbLogin(int&);
void mbLoginMenu(int&);
void mbbuyticket(int, int, double&);
void print_seating_chart();
double calculate_total_cost(int, int);
bool reserve_seat(int, int);
void mbwallet(double&);
void mbcancelacc(int, bool&);

int main() {

    admb();

    cin.get();
    return 0;
}

//big menu
void admb() {

    int choiceA, num;
    bool choice = false;
    char quitChoice;

    do {
        system("color 0d");
        cout << "\t|-----------------|" << endl;
        cout << "\t|  YSLL Cinema    |" << endl;
```

```
        cout << "\t|-----------------|" << endl;
        cout << "\t|1. Admin Mode    |" << endl;
        cout << "\t|2. Member Mode   |" << endl;
        cout << "\t|3. Quit          |" << endl;
        cout << "\t|-----------------|" << endl;
        cout << "     Please input a selection: ";
        cin >> choiceA;

        if (choiceA == 1) { //admin mode
            do {
                system("cls");
                admMenu(num);
                if (num == 1) { //movie list
                    read();
                }
                else if (num == 2) { //add movie
                    add();
                }
                else if (num == 3) { //delete movie
                    dlt();
                }
                else if (num > 4 || num < 0) { //num invalid
                    cout << "   Invalid input. Please press ENTER to
select again!" << endl;
                    cin.get();
                    cin.get();
                }
                else if (num == 4) //back
                    choice = false;
            } while (num > 0 && num != 4);
            system("cls");
        }

        else if (choiceA == 2) { //member mode
```

```
            system("cls");
            mbMenu();
        }
        else if (choiceA == 3) { //quit
            cout << "Confirm to QUIT the system?(Y/N): ";
            cin >> quitChoice;
            while (quitChoice != 'Y' && quitChoice != 'y' && quitChoice != 'n' && quitChoice != 'N') { //only accept Y|y or N|n
                cout << "Invalid input. Please re-type again! (Y-Yes, N-No): ";
                cin >> quitChoice;
            }
            if(quitChoice == 'Y' || quitChoice == 'y') //confirm to quit system
                choice = true;
            system("cls");
        }
        else { //choiceA invalid
            cout << "     Invalid input. Please press ENTER to select again!";
            cin.get();
            cin.get();
            system("cls");
        }
    } while (!choice);
}

//member sign in OR sign up
void mbMenu() {

    int choiceB, a; //'a' value will initialize in mbLogin function
    bool choice = false;

    do {
        system("color 0b");
```

```
        cout << "\t|--------------------------------------|" << endl;
        cout << "\t|  YSLL Cinema - NEW USER OR EXIST?    |" << endl;
        cout << "\t|--------------------------------------|" << endl;
        cout << "\t|1. New Register                       |" << endl;
        cout << "\t|2. User Login                         |" << endl;
        cout << "\t|--------------------------------------|" << endl;

        cout << "    Please input a selection: ";
        cin >> choiceB;

        if (choiceB == 1) { //register
            system("cls");
            mbSignup();
            system("cls");
        }

        else if (choiceB == 2) { //user login
            system("cls");
            mbLogin(a); //'a' value return by reference in mbLogin
function
            mbLoginMenu(a); //'a' value bring into mbLoginMenu function
            choice = true;
        }

        else { //choiceB invalid
            cout << "Invalid input. Please press ENTER to select again!"
<< endl;
            cin.get();
            cin.get();
            system("cls");
        }

    } while (!choice);
```

```
}

//member sign up
void mbSignup() {

    int i = 0;
    bool valid = false;
    memberSign member[arraySize];
    ofstream outfile("member.txt", ios::app);
    ifstream infile("member.txt", ios::app);

    while (!infile.eof()) { //read in member file
        getline(infile, member[i].name);
        infile.getline(member[i].password, pwSize);
        i++;
    }
    infile.close();
    //'i' now updated to the new array position

    system("color 0b");
    cout << "  ------------------------------------------------------------
----------------------------------" << endl;
    cout << "            ___    _______       _______________
_______                 ________    " << endl;
    cout << "           /      /      /      /             /         /
/                /      /    " << endl;
    cout << "          /      /      /      /             /         /
/                /      /     " << endl;
    cout << "         /      /      /      /      --------         /
/                /      /      " << endl;
    cout << "        /      /      /      /     /                 /      /
/      /       " << endl;
    cout << "       /             /      /      ----------       /      /
/      /        " << endl;
    cout << "      |_______      /                      /      /      /
/      /         " << endl;
```

```
    cout << "            /      /            -----      /      /      /
/      /           " << endl;

    cout << "          __/      /                /     /      /
_________       /       _________   " << endl;

    cout << "      _/          /        ---------      /      /
/      /               /   " << endl;

    cout << "    /           /      /               /      /
/      /               /    " << endl;

    cout << "   |___________/        _______________
/_______________/      /_______________/     " << endl;

    cout << "
_____________________________________________________________________________
_____________________" << endl;

    cout << endl;

    cout << "\t\t\t\t\tWelcome To YSLL Cinema!" << endl;

    cout << "\t\t\t\t\t  New Member Sign Up" << endl;

    cout << "\t\t   \033[31m*Please ensure that you set the PASSWORD
should be memorable!*\033[0m" << endl;

    cout << endl;


    bool usernamevalid = false;

    int a;


    cout << "\t\t\t\t\tUsername: ";

    cin.ignore();

    getline(cin, member[i].name); //input username


    do{

        for (a = 0; a < i; a++) { //check username not repeat then proceed
to set password

            if (member[i].name != member[a].name)

                usernamevalid = true;

        }


        for (a = 0; a < i; a++) { //check username repeat or not, if
repeat, cannot use second time
```

```
            if(member[i].name == member[a].name) { //a=3 i=4
                cout << "\t\t\t   \033[031mThis username has been created
by other user.\033[0m" << endl;
                cout << "\t\t\t\t\033[031mPlease re-type other
username!\033[0m" << endl;
                cout << endl;
                cout << "\t\t\t\t\tUsername: ";
                getline(cin, member[i].name);
                cin.ignore();
                usernamevalid = false;
            }
        }
    } while (!usernamevalid);


    outfile << endl;
    outfile << member[i].name << endl; //username save in file
    cout << endl;
    do {
        cout << "\t\t\t\t\t\033[34m*L-Letter; N-Number*\033[0m" << endl;
        cout << "\t\t\t\t\t\033[34m(Format - LLLLNNNN)\033[0m" << endl;
        cout << "\t\t\t\t\tPassword: ";
        cin.getline(member[i].password, pwSize); //input password


        if (testNum(member[i].password, pwSize)) { //identify password
format using bool function
            outfile << member[i].password; //password save in file
            cout << endl;
            cout << "\t\t\tMember sucessfully created! Your member ID is "
<< "\033[32m" << member[i].name << "0" << i+1 << "\033[0m" << "." << endl;
            cout << "\t\t\t\t  Press ENTER to return back to menu.";
            cin.get();
            valid = true;
        }
        else { //wrong password format
            cout << endl;
```

```
            cout << "\t\t\t\033[031mFormat incorrect! Please set the
PASSWORD again.\033[0m" << endl;
        }
    } while (!valid);

    outfile.close();
}

//identify password format
bool testNum(char memberpw[], int pwSize) {

    int count;

    for (count = 0; count < 4; count++) {
        if (!isalpha(memberpw[count]))
            return false;
    }

    for (count = 4; count < pwSize - 1; count++) {
        if (!isdigit(memberpw[count]))
            return false;
    }

    return true;
}

//member login
void mbLogin(int& a) { //initialize 'a' same as the variable 'lgno'
(member login)

    a = 0; //'a' define as the array location of current user login
    int i = 0, lgno = 0;
    bool valid = false;
    memberSign member[arraySize], loginmb[arraySize];
```

```
    ifstream infile("member.txt", ios::app);

    while (!infile.eof()) {
        getline(infile, member[i].name);
        infile.getline(member[i].password, pwSize);
        i++; //'i' can define as the total of existing member
    }
    infile.close();

    do {
        system("color 0b");
        cout << "  ------------------------------------------------------
------------------------------------" << endl;
        cout << "            ___    _______      _______________
________                 ________    " << endl;
        cout << "           /      /      /      /             /         /
/                /      /    " << endl;
        cout << "          /      /      /      /             /         /
/                /      /     " << endl;
        cout << "         /      /      /      /      --------         /
/                /      /      " << endl;
        cout << "        /      /      /      /     /                /
/                /      /       " << endl;
        cout << "       /             /      /      ----------       /
/                /      /        " << endl;
        cout << "      |_______      /                       /      /
/                /      /         " << endl;
        cout << "            /      /            -----      /      /
/                /      /          " << endl;
        cout << "         __/      /                 /     /      /
_________       /       _________   " << endl;
        cout << "       _/         /        ---------      /      /
/      /               /   " << endl;
        cout << "    /           /      /               /      /
/      /               /    " << endl;
        cout << "   |___________/        _______________
/________________/      /________________/     " << endl;
```

```
        cout << "
_____
_____" << endl;
        cout << endl;
        cout << "\t\t\t\t     Welcome To YSLL Cinema!" << endl;
        cout << "\t\t\t\t\t  Member Sign In" << endl;
        cout << endl;
        cin.ignore();
        cout << "\t\t\t\t\tUsername: ";
        getline(cin, loginmb[lgno].name); //input login name
        cout << endl;
        cout << "\t\t\t\t\tPassword: ";
        cin.getline(loginmb[lgno].password, pwSize); //input login
password


        for (a = 0; a < i; a++) { //identify whether the login name and
password exist inside the member file or not


            if ((member[a].name == loginmb[lgno].name) &&
(strcmp(member[a].password, loginmb[lgno].password) == 0)) { //name and
password also same, exist acc
                cout << "\t\t\tYou're successfully log in! Press ENTER to
MEMBER MENU." << endl;
                cin.get();
                system("cls");
                valid = true;
                break; //once find the exist acc, get inside IF function,
need to stop the FOR loop to invoid execute a++
            }


        }
        if ((member[a].name != loginmb[lgno].name) ||
(strcmp(member[a].password, loginmb[lgno].password) == -1)) { //invalid
account
            cout << "\t\t\t    \033[31mNOT existing account, press ENTER
to sign in again.\033[0m" << endl;
            cin.get();
            system("cls");
```

```
        }
    } while (!valid);

}


//member login menu
void mbLoginMenu(int& a) {

    int choiceC, current_idno = a; //'current_idno' initialize to 'a'
    double balance = 0; //declare member wallet balance
    movieType movie[arraySize];
    bool loginChoice = false, auto_logout = false, buyticketChoice =
false;

    do {
        system("color 0b");
        cout << "\t|--------------------------------------|" << endl;
        cout << "\t|  YSLL Cinema - Member Login Menu     |" << endl;
        cout << "\t|--------------------------------------|" << endl;
        cout << "\t|1. Movie List                         |" << endl;
        cout << "\t|2. Buy Ticket                         |" << endl;
        cout << "\t|3. Wallet                             |" << endl;
        cout << "\t|4. Delete Account                     |" << endl;
        cout << "\t|5. Logout                             |" << endl;
        cout << "\t|--------------------------------------|" << endl;
        cout << "\tPlease input a selection: ";
        cin >> choiceC;
        cout << endl;

        if (choiceC == 1) { //movie list
            read();
        }
```

```
        else if (choiceC == 2) { //buy ticket
            do {
                system("cls");
                ifstream infile("cinema.txt", ios::app);
                int i = 0, movieNo;

                cout << "\t1. Buy Ticket" << endl;
                cout << "\t|-----------------------------------------------
---------|" << endl;
                cout << "\t|\t\t\tMovie List\t\t\t|" << endl;
                cout << "\t|-----------------------------------------------
---------|" << endl;
                cout << "\t| Name\t\t\t\t\t      | Price   |" << endl;
                cout << "\t|-----------------------------------------------
---------|" << endl;

                if (infile.is_open()) {
                    while (!infile.eof()) { //read in file
                        getline(infile, movie[i].name);
                        infile >> movie[i].price;
                        cout << "\t| " << i + 1 << ". " << movie[i].name
<< endl;
                        cout << fixed << setprecision(2);
                        cout << "\t|\t\t\t\t\t      | RM" <<
movie[i].price << " |" << endl;
                        i++; //total movie number
                        infile.ignore();
                    }
                    cout << "\t|-----------------------------------------
-------------|" << endl;
                    infile.close();
                }

                cout << "\tChoose a movie: ";
                cin >> movieNo;
```

```
                if (movieNo <= i) { //choose movieNo within the movie
number
                    mbbuyticket(movieNo, i, balance); //balance renew
                    system("cls");
                    buyticketChoice = true;
                }
                else { //movieNo chosen out of bound
                    cout << "\tInvalid input. Press ENTER to re-select
again!" << endl;
                    cin.get();
                    cin.get();
                }
            } while (!buyticketChoice);


        }
        else if (choiceC == 3) { //wallet
            mbwallet(balance);
            system("cls");
        }
        else if (choiceC == 4) { //delete account
            mbcancelacc(current_idno, auto_logout); //'current_idno' = 'a'
value bring into mbcancelacc function
            system("cls");
            if (auto_logout) //'auto_logout' value takeout from
mbcancelacc function to do IF function
                loginChoice = true; //quit from this whole loop
        }
        else if (choiceC == 5) { //logout
            cout << "\tThank for your support to our YSLL Cinema!" <<
endl;
            cout << "\tPress ENTER to proceed logout." << endl;
            cin.get();
            cin.get();
            system("cls");
            loginChoice = true;
```

```
        }

    } while(!loginChoice);
}

//member buy ticket
void mbbuyticket(int movieNo, int i, double& balance) { //i = total movie number

    movieType movie[arraySize];
    system("cls");
    system("color 0d");

    int date, time, numSeats = 0;
    double payment, totalPayment, change;

    while (true) {
        cout << "\t------------------------------" << endl;
        cout << "\t|       Date Available        |" << endl;
        cout << "\t------------------------------" << endl;
        cout << "\t|        1.16/4/2023          |" << endl;
        cout << "\t|        2.17/4/2023          |" << endl;
        cout << "\t|        3.18/4/2023          |" << endl;
        cout << "\t|        4.19/4/2023          |" << endl;
        cout << "\t|        5.20/4/2023          |" << endl;
        cout << "\t------------------------------" << endl;
        cout << "\tSelect a date: ";
        cin >> date;

        if (date >= 1 && date <= 5) {
            break;
        }
        else {
            cout << "Please enter a number between 1 to 5 only\n\n";
```

```
    }
}
cout << endl;
system("CLS");

while (true) {
    cout << "You have selected the date" << date << "." << endl;
    switch (date) {
    case 1:
        cout << "16/4/2023";
        break;
    case 2:
        cout << "17/4/2023";
        break;
    case 3:
        cout << "18/4/2023";
        break;
    case 4:
        cout << "19/4/2023";
    case 5:
        cout << "20/4/2023";
        break;
    }
    cout << endl;

    cout << "Do you want to change your selection? (Y/N): ";
    char choice;
    cin >> choice;
    system("CLS");

    if (choice == 'Y' || choice == 'y') {
        while (true) {
            cout << "\t-----------------------------" << endl;
```

```
                cout << "\t|       Date Available        |" << endl;
                cout << "\t-----------------------------" << endl;
                cout << "\t|        1.16/4/2023          |" << endl;
                cout << "\t|        2.17/4/2023          |" << endl;
                cout << "\t|        3.18/4/2023          |" << endl;
                cout << "\t|        4.19/4/2023          |" << endl;
                cout << "\t|        5.20/4/2023          |" << endl;
                cout << "\t-------------------------------" << endl;
                cout << "\tSelect a date: ";
                cin >> date;

                if (date >= 1 && date <= 5) {
                    break;
                }
                else {
                    cout << "Please enter a number between 1 to 5 only" <<
endl << endl;
                }
            }
            cout << endl;
            system("CLS");
        }
        else if (choice == 'N' || choice == 'n') {
            break;
        }
        else {
            cout << "Invalid input. Please enter Y or N only." << endl <<
endl;
        }
    }

    cout << "You have selected date " << date << "." << endl;
    switch (date) {
    case 1:
```

```
    cout << "16/4/2023";
    break;
case 2:
    cout << "17/4/2023";
    break;
case 3:
    cout << "18/4/2023";
    break;
case 4:
    cout << "19/4/2023";
case 5:
    cout << "20/4/2023";
    break;
}
cout << endl;
system("CLS");

//Choosing time section
system("color 0e");
while (true) {
    cout << "\t------------------------------" << endl;
    cout << "\t|         Time available     |" << endl;
    cout << "\t------------------------------" << endl;
    cout << "\t|         1. 10:00 AM        |" << endl;
    cout << "\t|         2. 01:15 PM        |" << endl;
    cout << "\t|         3. 01:45 PM        |" << endl;
    cout << "\t|         4. 03:45 PM        |" << endl;
    cout << "\t|         5. 04:15 PM        |" << endl;
    cout << "\t|         6. 06:45 PM        |" << endl;
    cout << "\t|         7. 09:15 PM        |" << endl;
    cout << "\t------------------------------" << endl;
    cout << "\tSelect a time: ";
    cin >> time;
```

```
    if (time >= 1 && time <= 7) {
        break;
    }
    else {
        cout << "Please enter a number between 1 to 7 only\n\n";
    }
}
cout << endl;
system("CLS");

while (true) {
    cout << "You have selected the time" << time << "." << endl;
    switch (time) {
    case 1:
        cout << "10:00 AM";
        break;
    case 2:
        cout << "01:15 PM";
        break;
    case 3:
        cout << "01:45 PM";
        break;
    case 4:
        cout << "03:45 PM";
        break;
    case 5:
        cout << "04:15 PM";
        break;
    case 6:
        cout << "06:45 PM";
        break;
    case 7:
        cout << "09:15 PM";
```

```
            break;
        }
        cout << endl;

        cout << "Do you want to change your selection? (Y/N): ";
        char choice;
        cin >> choice;
        system("CLS");

        if (choice == 'Y' || choice == 'y') {
            while (true) {
                cout << "\t------------------------------" << endl;
                cout << "\t|         Time available     |" << endl;
                cout << "\t------------------------------" << endl;
                cout << "\t|         1. 10:00 AM        |" << endl;
                cout << "\t|         2. 01:15 PM        |" << endl;
                cout << "\t|         3. 01:45 PM        |" << endl;
                cout << "\t|         4. 03:45 PM        |" << endl;
                cout << "\t|         5. 04:15 PM        |" << endl;
                cout << "\t|         6. 06:45 PM        |" << endl;
                cout << "\t|         7. 09:15 PM        |" << endl;
                cout << "\t------------------------------" << endl;
                cout << "\tSelect a time: ";
                cin >> time;

                if (time >= 1 && time <= 7) {
                    break;
                }
                else {
                    cout << "Please enter a number between 1 to 7 only" <<
endl << endl;
                }
            }
            cout << endl;
```

```
            system("CLS");
        }
        else if (choice == 'N' || choice == 'n') {
            break;
        }
        else {
            cout << "Invalid input. Please enter Y or N only." << endl << endl;
        }
    }

    cout << "You have selected time " << time << "." << endl;
    switch (time) {
    case 1:
        cout << "10.00 AM";
        break;
    case 2:
        cout << "01:15 PM";
        break;
    case 3:
        cout << "01:45 PM";
        break;
    case 4:
        cout << "03:45 PM";
        break;
    case 5:
        cout << "04:15 PM";
        break;
    case 6:
        cout << "06:45 PM";
        break;
    case 7:
        cout << "09:15 PM";
        break;
```

```
    }
    cout << endl;
    system("CLS");

    cout << endl;

    char choice = 'y';
    do {
        // print seating chart
        system("color 0e");
        cout << "\t\t----------------------------------------" << endl;
        cout << "\t\t|                                      |" << endl;
        cout << "\t\t----------------------------------------" << endl;
        cout << "\t\t\t\tSCREEN\n" << endl;

        print_seating_chart();

        // get user input
        cout << endl;
        cout << "\t\tEnter column number (1-" << columns << "): ";
        int seat;
        cin >> seat;
        cout << "\t\tEnter row number (1-" << rows << "): ";
        int row;
        cin >> row;

        // validate input
        if (row < 1 || row > rows || seat < 1 || seat > columns) {
            cout << "\t\tInvalid input. Press ENTER to select again." <<
endl;
            cin.get();
            cin.get();
            system("cls");
            continue; //skip below and loop again
```

```
        }

        // reserve seat
        bool reserved = reserve_seat(row, seat);
        if (reserved) {
            cout << "\t\tSeat reserved." << endl;
            numSeats++;
        }
        else {
            cout << "\t\tSeat is already reserved. Press ENTER to select
again!" << endl;
            cout << endl;
            cin.get();
            cin.get();
            system("cls");
            continue;
        }

        // ask whether the user wants to reserve more seats or not
        cout << "\t\tDo you want to reserve more seats? (Y/N): ";
        cin >> choice;
        system("cls");
    } while (choice == 'Y' || choice == 'y');



    // calculate total cost
    totalPayment = calculate_total_cost(numSeats, movieNo);

    // display the total payment for the user
    cout << endl << endl;
    cout << "\t\tTotal price is: RM" << totalPayment << endl;
    cout << "\t\tWallet balance is: RM" << balance << endl;
```

```
    // calculate change for the extra payment
    change = balance - totalPayment;
    bool payment_successful = true;

    // output change of extra payment or error message
    if (change >= 0) {
        balance = change;
        cout << "\t\tChange: RM" << change << endl;
        cout << "\t\tThank you for purchasing and have a good day :)!" << endl;
        cout << endl;
    }
    else {
        cout << "\t\t\033[31mError: Insufficient amount.\033[0m" << endl;
        cout << "\t\t\033[31mPayment Unsuccessful, please top up WALLET before transaction.\033[0m" << endl;
        cout << "\t\t\033[31mPress ENTER return back to Member Menu.\033[0m" << endl;
        cin.get();
        cin.get();
        payment_successful = false;
    }

    cout << endl;



    while (payment_successful) {
        cout << "\t\t-------------------------------------------------" << endl;
        cout << "\t\t*              YSLL Cinema Receipt              *" << endl;
        cout << "\t\t-------------------------------------------------" << endl;
        cout << "\t\t*Movie Title: ";

        ifstream infile("cinema.txt");
```

```
while (!infile.eof()) {
    for (int a = 0; a < i; a++) {
        getline(infile, movie[a].name);
        infile >> movie[a].price;
        infile.ignore();
    }
}
infile.close();

cout << movie[movieNo - 1].name;

cout << endl;

cout << "\t\t*Movie Date: ";
switch (date) {
case 1:
    cout << "16/4/2023";
    break;
case 2:
    cout << "17/4/2023";
    break;
case 3:
    cout << "18/4/2023";
    break;
case 4:
    cout << "19/4/2023";
case 5:
    cout << "20/4/2023";
    break;
}
cout << endl;

cout << "\t\t*Movie Time: ";
```

```
        switch (time) {
        case 1:
            cout << "10.00 AM";
            break;
        case 2:
            cout << "01:15 PM";
            break;
        case 3:
            cout << "01:45 PM";
            break;
        case 4:
            cout << "03:45 PM";
            break;
        case 5:
            cout << "04:15 PM";
            break;
        case 6:
            cout << "06:45 PM";
            break;
        case 7:
            cout << "09:15 PM";
            break;
        }
        cout << endl;


        cout << "\t\t|\t\t\t\t\t\t|" << endl;
        cout << "\t\t|" << "Total payment:RM" << totalPayment <<
"\t\t\t\t|" << endl;
        cout << "\t\t|\t\t\t\t\t\t|" << endl;
        cout << "\t\t|Show this ticket to the counter to clarify\t| " <<
endl;
        cout << "\t\t|the ticket validity due to security reason\t|" <<
endl;
        cout << "\t\t|then only assign according to the seating\t|" <<
endl;
```

```
        cout << "\t\t|postion. Sorry for problem caused.\t\t|" << endl;
        cout << "\t\t|\t\t\t\t\t\t|" << endl;
        cout << "\t\t|\t\t\t\t\t\t|" << endl;
        cout << "\t\t|Have a great day! :)\t\tQR CODE\t\t|" << endl;
        cout << "\t\t|\t\t\t*********\t|" << endl;
        cout << "\t\t|\t\t\t*\t*\t|" << endl;
        cout << "\t\t|\t\t\t*\t*\t|" << endl;
        cout << "\t\t|\t\t\t*********\t|" << endl;
        cout << "\t\t|\t\t\t\t\t\t|" << endl;
        cout << "\t\t-------------------------------------------------" <<
endl;

        cin.get();
        cin.get();
        payment_successful = false;
    }
}

//choose cinema seat
void print_seating_chart() {

    cout << endl;
    for (int i = 0; i < rows; i++) {
        cout << "\t\t";
        for (int j = 0; j < columns; j++) {
            if (seats[i][j] == 0) {
                cout << "O ";
            }
            else {
                cout << "X ";
            }
        }
        cout << endl;
    }
```

```
}

//calculate total payment of the chosen movie with chosen how many seats
double calculate_total_cost(int numSeats, int movieNo) {
    movieType movie[arraySize];
    ifstream infile("cinema.txt");
    int i = 0;
    while (!infile.eof()) {
        infile.ignore();
        getline(infile, movie[i].name);
        infile >> movie[i].price;
        i++;
    }
    return numSeats * movie[movieNo-1].price;
}

//chosen seat reserved
bool reserve_seat(int row, int seat) {
    if (seats[row - 1][seat - 1] == 0) {
        seats[row - 1][seat - 1] = 1;
        return true;
    }
    else {
        return false;
    }
}

//member wallet
void mbwallet(double& balance) {

    int topup;
    char topupChoice;
    bool topupexit = false;
```

```
    system("cls");
    cout << "\t|----------------------------|" << endl;
    cout << "\t|           Wallet           |" << endl;
    cout << "\t|----------------------------|" << endl;
    cout << fixed << setprecision(2);
    cout << "\t     Balance = RM" << balance << endl;
    cout << "\t|----------------------------|" << endl;
    cout << "\tWould you like to top up your wallet?(Y/N): ";
    cin >> topupChoice;

    while (topupChoice != 'Y' && topupChoice != 'y' && topupChoice != 'N'
&& topupChoice != 'n'){
        cout << "\tInvalid input. Please re-type again! (Y-Yes, N-No): ";
        cin >> topupChoice;
    }

    if (topupChoice == 'Y' || topupChoice == 'y') {

        system("cls");
        cout << "\t|----------------|" << endl;
        cout << "\t| Top Up Service |" << endl;
        cout << "\t|----------------|" << endl;
        cout << "\t|      RM10      |" << endl;
        cout << "\t|      RM20      |" << endl;
        cout << "\t|      RM50      |" << endl;
        cout << "\t|      RM100     |" << endl;
        cout << "\t|----------------|" << endl;
        do {
            cout << endl;
            cout << "\tPlease key in a value to TOPUP: ";
            cin >> topup;

            switch (topup) {
```

```
            case 10: {
                balance += topup;
                break;
            }
            case 20: {
                balance += topup;
                break;
            }
            case 50: {
                balance += topup;
                break;
            }
            case 100: {
                balance += topup;
                break;
            }
            default: {
                cout << "\tInvalid input. Please re-type TOPUP value!" <<
endl;
                continue;
            }
            }

            cout << "\tWould you like to TOP UP again? (Y/N): ";
            cin >> topupChoice;

            while (topupChoice != 'Y' && topupChoice != 'y' &&
topupChoice != 'N' && topupChoice != 'n') {
                cout << "\tInvalid input. Please re-type again! (Y-Yes, N-
No): ";
                cin >> topupChoice;
            }
            //if (topupChoice == 'Y' || topupChoice == 'y') loop again
```

```
            if (topupChoice == 'N' || topupChoice == 'n') {
                cout << "\tPress ENTER to return back to Member MENU." <<
endl;
                cin.get();
                cin.get();
                topupexit = true;
            }

        } while (!topupexit);

    }

    else if (topupChoice == 'N' || topupChoice == 'n') {
        cout << "\tPress ENTER to return back to Member MENU." << endl;
        cin.get();
        cin.get();
    }
}

//member delete account
void mbcancelacc(int current_idno, bool& auto_logout) { //'current_idno'
come from 'a'

    int i = 0;
    char deleteChoice;
    memberSign member[arraySize];
    ifstream infile("member.txt", ios::app);

    while (!infile.eof()) {
        getline(infile, member[i].name);
        infile.getline(member[i].password, pwSize);
        i++;
    }
    infile.close();
```

```
    cout << "\tDo you want to delete your current account? (Y/N): ";
    cin >> deleteChoice;

    while (deleteChoice != 'Y' && deleteChoice != 'y' && deleteChoice != 
'N' && deleteChoice != 'n') {
        cout << "\t\033[031mInvalid input. Please re-type again! (Y-Yes, 
N-No): \033[0m";
        cin >> deleteChoice;
    }

    if (deleteChoice == 'Y' || deleteChoice == 'y') {

        ofstream outfile("member.txt", ios::trunc); //rewrite the file

        for (int a = 0; a < i-1; a++) {
            if ((member[a].name == member[current_idno].name) && 
(strcmp(member[a].password,member[current_idno].password) == 0)) {

                member[a].name = member[a + 1].name; //replace
                strcpy_s(member[a].password, member[a + 1].password); 
//replace

                if (a != 0)
                    outfile << endl;
                outfile << member[a].name << endl; //write in file
                outfile << member[a].password; //write in file
            }

            else if ((member[a].name != member[current_idno].name) || 
(strcmp(member[a].password, member[current_idno].password) == -1)) {

                if (a != 0)
                    outfile << endl;
                outfile << member[a].name << endl; //write back to file
```

```
                outfile << member[a].password; //write back to file
            }
        }
        outfile.close();
        cout << "\tAccount deleted. Press ENTER auto logout..." << endl;
        cin.get();
        cin.get();
        auto_logout = true;
    }
    else if (deleteChoice == 'N' || deleteChoice == 'n') {
        cout << "\tPress ENTER return back to Member MENU." << endl;
        cin.get();
        cin.get();
    }


}

//admin menu
void admMenu(int& choice) {

    system("color 0a");
    cout << "\t|--------------------------------------|" << endl;
    cout << "\t|  YSLL Cinema - Admin Operating Menu  |" << endl;
    cout << "\t|--------------------------------------|" << endl;
    cout << "\t|1. Movie List                         |" << endl;
    cout << "\t|2. Add Movie                          |" << endl;
    cout << "\t|3. Delete Movie                       |" << endl;
    cout << "\t|4. Back                               |" << endl;
    cout << "\t|--------------------------------------|" << endl;
    cout << "\t    Please input a selection: ";
    cin >> choice;
}
```

```
//view movie
void read() {


    ifstream infile("cinema.txt", ios::app);
    movieType movie[arraySize];
    int i = 0;


    system("cls");
    cout << "\t|-----------------------------------------------------|"
<< endl;
    cout << "\t|\t\t\tMovie List\t\t\t|" << endl;
    cout << "\t|-----------------------------------------------------|"
<< endl;
    cout << "\t| Name\t\t\t\t\t      | Price   |" << endl;
    cout << "\t|-----------------------------------------------------|"
<< endl;


    if (infile.is_open()) {
        while (!infile.eof()) {
            getline(infile, movie[i].name);
            infile >> movie[i].price;
            cout << "\t| " << i+1 << ". " << movie[i].name << endl;
            cout << fixed << setprecision(2);
            cout << "\t|\t\t\t\t\t      | RM" << movie[i].price << " |" <<
endl;
            i++;
            infile.ignore();
        }
        cout << "\t|-----------------------------------------------------
-|" << endl;
        infile.close();
    }
    else
        cout << "\tUnable to open storing file!" << endl;
```

```
    cout << "\tPress ENTER to return back to Operating Menu." << endl;
    cin.get();
    cin.get();
    system("cls");
}

//add movie
void add() {

    ifstream infile("cinema.txt", ios::app);
    ofstream outfile("cinema.txt", ios::app);
    movieType movie[arraySize];
    char addMovie;

    do {
        int i = 0;
        while (!infile.eof()) {
            getline(infile, movie[i].name);
            infile >> movie[i].price;
            i++;
            infile.ignore();
        }
        cout << endl;
        cout << "\tAdd Movie" << endl;
        cout << "\tName: ";
        outfile << endl;
        cin.ignore();
        getline(cin, movie[i].name);
        outfile << movie[i].name << endl;
        cout << "\tPrice: ";
        cin >> movie[i].price;
        outfile << movie[i].price;
```

```
        cout << "\tDo you want to add more movie?(Y/N): ";
        cin >> addMovie;

        while (addMovie != 'Y' && addMovie != 'y' && addMovie != 'N' &&
addMovie != 'n') {
            cout << "\tInvalid input. Please type again (Y-Yes, N-No): ";
            cin >> addMovie;
        }

    } while (addMovie == 'Y' || addMovie == 'y');

    infile.close();
    outfile.close();

    cout << endl;
    cout << "\tPress ENTER to return back to Operating Menu." << endl;
    cin.get();
    cin.get();
    system("cls");
}

//delete movie
void dlt() {
    movieType movie[arraySize];
    int i, a = 0, dlno;
    bool choice = false;

    do {
        ifstream infile("cinema.txt", ios::app);
        i = 0;
        system("cls");

        cout << "\tPlease choose a movie to delete." << endl;
```

```
        cout << "\t|-----------------------------------------------------
-|" << endl;
        cout << "\t|\t\t\tMovie List\t\t\t|" << endl;
        cout << "\t|-----------------------------------------------------
-|" << endl;
        cout << "\t| Name\t\t\t\t\t      | Price   |" << endl;
        cout << "\t|-----------------------------------------------------
-|" << endl;


        if (infile.is_open()) {
            while (!infile.eof()) {
                getline(infile, movie[i].name);
                infile >> movie[i].price;
                cout << "\t| " << i + 1 << ". " << movie[i].name << endl;
                cout << fixed << setprecision(2);
                cout << "\t|\t\t\t\t\t      | RM" << movie[i].price << "
|" << endl;
                i++;
                infile.ignore();
            }
            cout << "\t|-----------------------------------------------------
-----|" << endl;
            cout << endl;
            infile.close();
        }
        else
            cout << "Unable to open storing file!" << endl;
        //i is the total number of movie
        //i - 1 is the index of last movie
        cout << "\tWhich one you want to delete? (Type MovieNo to
Delete) : ";
        cin >> dlno;


        if (dlno <= i && dlno > 0)
            choice = true;
```

```
        else if (dlno > i || dlno <= 0) {
            cout << "\tInvalid input. Press ENTER to select again!" <<
endl;
            cin.get();
            cin.get();
        }
    } while (!choice);

    dlno -= 1; //dlno array or index position

    ofstream outfile("cinema.txt", ios::trunc); //rewrite the file

    while (a < i - 1) { //delete movie will be deducted from file, index
also -1

        int pos = movie[a].name.find(movie[dlno].name, 0);

        if (pos != string::npos) {

            while (a < i - 1) {

                movie[a].name = movie[a + 1].name;
                movie[a].price = movie[a + 1].price;

                if (a != 0)
                    outfile << endl;
                outfile << movie[a].name << endl;
                outfile << movie[a].price;
                pos = movie[a].name.find(movie[dlno].name, pos +
movie[dlno].name.length());
                a++;
            }
        }
```

```
        else if (pos == string::npos && (a != i - 1)) {
            if (a != 0)
                outfile << endl;
            outfile << movie[a].name << endl;
            outfile << movie[a].price;
            a++;
        }
    }

    outfile.close();
    cout << "\tPress ENTER to return back to Operating Menu." << endl;
    cin.get();
    cin.get();
    system("cls");
}
```